Problem Set 9: Pricing Securities in a Finite-State Markov Economy

Economics 5040: Dynamic Stochastic Modeling

Fall 2026

Overview

This problem set develops a general method for pricing state-contingent securities at arbitrary horizons in a finite-state Markov economy. With CRRA utility,

$$u'(c) = c^{-\rho}, \qquad m_{ij} = \beta \left(\frac{c_j}{c_i} \right)^{-\rho},$$

and the one-period pricing matrix is

$$Q_{ij} = P_{ij} m_{ij}.$$

If a security pays the state-contingent payoff vector r exactly T periods from today, its price vector is

$$\boxed{p^{(T)} = Q^T r}.$$

The Economy

Unless otherwise stated, use

$$\beta = 0.97, \qquad \rho = 2, \qquad c = \begin{pmatrix} 0.5 \\ 1.0 \\ 1.5 \end{pmatrix},$$

and

$$P = \begin{pmatrix} 0.80 & 0.15 & 0.05 \\ 0.10 & 0.80 & 0.10 \\ 0.05 & 0.15 & 0.80 \end{pmatrix}.$$

Interpret the states as low, normal, and high aggregate consumption.

Problem 1: From Marginal Utility to State Prices

(a) Starting from $m_{ij} = \beta u'(c_j)/u'(c_i)$, derive

$$m_{ij} = \beta (c_j/c_i)^{-\rho}.$$

(b) Compute the complete 3×3 stochastic discount factor matrix $M = [m_{ij}]$.

(c) Explain why m_{ij} is high when the economy moves from high consumption today to low consumption tomorrow.

(d) Construct and report numerically $Q = P \circ M$.

(e) Interpret Q_{ij}. What security has price Q_{ij} when today's state is i?

(f) Price an Arrow security paying one unit next period only in the low-consumption state, for each possible current state.

Problem 2: An Arbitrary One-Period Security

Consider

$$r = \begin{pmatrix} 0.5 \\ 1.0 \\ 2.0 \end{pmatrix}.$$

(a) Write $p_i = \sum_{j=1}^{3} P_{ij} m_{ij} r_j$ and explain the role of each component.

(b) Show that $p = Qr$ and compute p.

(c) Now price

$$\tilde{r} = \begin{pmatrix} 2.0 \\ 1.0 \\ 0.5 \end{pmatrix}.$$

(d) The securities contain the same three payoff amounts but allocate them to different states. Compare prices and explain the difference using marginal utility.

(e) Explain why r is a payoff vector rather than a return vector.

Problem 3: Pricing at an Arbitrary Horizon

Consider an Arrow claim paying

$$r = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

exactly T periods from today.

(a) Show that $p^{(1)} = Qr$.

(b) Show that $p^{(2)} = Q^2 r$ and explain why the pricing operator is applied twice.

(c) Show by repeated substitution or induction that

$$\boxed{p^{(T)} = Q^T r}.$$

(d) Compute the price vector for $T = 1, 2, 5, 10, 20$ and present the results in a table.

(e) For each current state, plot the price for $T = 1, \ldots, 30$.

(f) Does the importance of the initial state diminish with maturity? Explain what you observe.

Problem 4: Zero-Coupon Bonds and the Term Structure

A T-period zero-coupon bond pays one unit in every state at $t + T$, so its payoff vector is $\mathbf{1}$.

(a) Show that

$$\boxed{b^{(T)} = Q^T \mathbf{1}}.$$

(b) Compute $b^{(1)}$ in each current state.

(c) Compute $R_i^f = 1/b_i^{(1)}$. Which state has the lowest risk-free rate? Explain.

(d) For $T = 1, \ldots, 30$, compute

$$y_i(T) = -\frac{1}{T} \log b_i^{(T)}.$$

(e) Plot the three yield curves on the same graph.

(f) Explain why a claim with a certain terminal payoff can have a price that depends on today's state.

Problem 5: Risk Aversion and Asset Prices

Return to the $T = 5$ low-consumption Arrow security. Hold $\beta = 0.97$ fixed.

(a) Compute its price vector for $\rho = 0, 1, 2, 5, 10$.

(b) Plot price against ρ for each current state.

(c) Show that when $\rho = 0$, $M_{ij} = \beta$ for all i, j.

(d) Explain why increasing ρ changes the value of a claim paying in the low-consumption state.

(e) Repeat for a claim paying only in the high-consumption state,

$$\tilde{r} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix},$$

and compare the results.

(f) Interpret: "An asset is valuable not merely because it pays a lot, but because of when it pays."

Problem 6: Patience, Maturity, and Discounting

Hold $\rho = 2$ fixed and consider $\beta = 0.90, 0.95, 0.97, 0.99$.

(a) For each β, compute zero-coupon bond prices for $T = 1, 5, 10, 20$.

(b) Explain why increasing β raises asset prices.

(c) Is the percentage effect of changing β larger for a one-period or twenty-period claim? Explain.

(d) Use Q^T to explain why small changes in one-period discounting can have large long-horizon effects.

Problem 7: Write a General Security-Pricing Function

Write

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price_security(rho, beta, P, c, r, T)
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that returns M, Q, and $p^{(T)}$. It should verify that P is square and nonnegative, its rows sum to one, c and r have compatible dimensions, consumption is strictly positive, and T is a nonnegative integer.

Use your function to price:

(a) a one-period Arrow security;

(b) a ten-period Arrow security;

(c) a five-period zero-coupon bond;

(d) $r = (1, 2, 4)'$ at $T = 1, 5, 10$.

Report the complete price vector in every case.

Problem 8: Synthesis

Answer each in a short paragraph.

(a) What are the distinct economic roles of P, M, and Q?

(b) Why is $Q^T r$ not simply the expected payoff discounted by the scalar β^T?

(c) If two securities have the same expected payoff under the physical probabilities, must they have the same price? Explain.

(d) What feature makes an asset a useful hedge against aggregate consumption risk?

(e) Explain, mathematically and economically, the meaning of

$$\boxed{p^{(T)} = Q^T r}.$$

Submission

Submit one Jupyter notebook containing all derivations, clearly organized Python code, numerical results, requested tables and figures, and economic interpretations in complete sentences. The notebook should run from beginning to end without errors.